

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
- Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
- All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
- Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
- Bonus problems are not required for credit, but they are worth extra credit.

Problems

Problem 1. (Gradient Descent for Linear Regression.) Consider the linear regression problem:

$$\min_{\theta} J(\theta) = \min_{\theta} \|X\theta - y\|^2, \quad (1)$$

where $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$. When the problem is overdetermined i.e. $n > d$ and X is full column-rank, we saw in Module 2 that $X^\top X$ is invertible and the solution to the problem in (1) is $\hat{\theta}_{\text{OD}} = (X^\top X)^{-1} X^\top y$.

In HW4 and in lecture, we considered the underdetermined case, i.e. $n < d$; in this case, the problem (1) has infinitely many solutions with objective value zero (i.e. such that $\|X\theta - y\| = 0$). If X is full row-rank, then $XX^\top$ is invertible and the minimum norm solution admits the closed form expression $\hat{\theta}_{\text{UD}} = X^\top (XX^\top)^{-1} y$.

In this problem, we will explore the solution found by gradient descent for problem (1) for both cases. Refer to the initial point of gradient descent as θ_0 and the iterates as $\{\theta_t\}_{t=1}^T$. We say that the algorithm converges if there exists a τ such that $\theta_t = \theta_\tau$ for all $t > \tau$, and we refer to $\hat{x}_{\text{GD}} = x_\tau$ as the solution obtained by gradient descent.

- Gradient descent update:** Compute the gradient of $J(\theta)$ and write the gradient descent update

$$\theta_{t+1} = \theta_t - \eta \nabla J(\theta_t)$$

explicitly in terms of data (X, y) , and step-size η .

Solution.

$$J(\theta) = \frac{1}{2} \|X\theta - y\|_2^2 \implies \nabla J(\theta) = X^\top (X\theta - y).$$

Hence for step size $\eta > 0$,

$$\theta_{t+1} = \theta_t - \eta X^\top (X\theta_t - y) = (I - \eta X^\top X)\theta_t + \eta X^\top y.$$

- b. **Fixed point condition:** Suppose that gradient descent converges to a point θ_∞ . Write down the equation it must satisfy. What is the interpretation of this condition?

Solution.

If $\theta_t \rightarrow \theta_\infty$, taking limits in the update (or setting $\nabla J(\theta_\infty) = 0$) gives

$$0 = X^\top (X\theta_\infty - y) \iff \boxed{X^\top X \theta_\infty = X^\top y.}$$

Interpretation: θ_∞ satisfies the *normal equations*, i.e., it is a least-squares solution (possibly non-unique when $X^\top X$ is singular).

- c. **Solution obtained by gradient descent:** Now suppose that we are in the overdetermined case, and assume full column-rank. Write down the solution obtained by gradient descent, assuming that training converges to θ_∞ .

Solution.

From the definition of convergence in the question, we just solve $\lim_{\tau \rightarrow \infty} \nabla_\theta J(\theta_\tau) = 0$ for θ_∞ . Hence, the solution obtained by gradient descent is exactly $\hat{\theta}_{\text{OD}} = (X^\top X)^{-1} X^\top y$.

Problem 2. (Gradient Descent for Linear Regression: Underdetermined Case.) Consider a simple special case to develop intuition: let $n = 1, d = 2$, and choose $X = \begin{bmatrix} 2 & 1 \end{bmatrix}$ and $y = [2]$.

- a. Show that there exist infinitely many $\hat{\theta}$ satisfying $X^\top \hat{\theta} = y$ on a line ℓ . Write the equation of the line ℓ .

Solution. the condition, $X^\top \hat{\theta} = y$, for zero objective value, we have

$$2\hat{\theta}_1 + 1\hat{\theta}_2 = 2 \implies \hat{\theta}_2 = 2 - 2\hat{\theta}_1$$

The above is the equation for line ℓ .

- b. Starting from $\theta_0 = 0$, what is the direction of the negative gradient, written as a unit-norm vector? Does the direction change along the trajectory?

Solution. First, we find the expression for the gradient

$$\nabla_\theta J(\theta) = 2X^\top X\theta - 2X^\top y = 10 \cdot \theta - 4X^\top = 10\theta - 4 \cdot \begin{bmatrix} 2 & 1 \end{bmatrix}^\top$$

Plugging in $\theta_0 = 0$, we have $-\nabla_\theta J(0) = 4 \cdot \begin{bmatrix} 2 & 1 \end{bmatrix}$. Normalizing, we have the direction is $u = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 2 & 1 \end{bmatrix}$. Hence $\theta_1 = C_1 u$ for some constant C_1 depending on our stepsize η . Then,

$$-\nabla_\theta J(\theta_1) = -10 \cdot C_1 u + 4\sqrt{5}u = (-10 \cdot C_1 + 4\sqrt{5})u$$

Thus, the direction of the negative gradient does not change at timestep $t = 2$ as well, and we can argue inductively that it will not change for future timesteps either.

- c. Now, we explore for the simple example which solution gradient descent finds. Assume that gradient descent converges (as defined in **Problem 1**).

- (i) Using your findings from the previous part, which solution $\hat{\theta}_{\text{GD}}$ on line ℓ does gradient descent find?
- (ii) Verify that for any other solution $\hat{\theta} \in \ell$, it holds that $(\hat{\theta} - \hat{\theta}_{\text{GD}}) \perp \hat{\theta}_{\text{GD}}$.

- (iii) Through a pictorial sketch, argue that $\hat{\theta}_{\text{GD}}$ has the smallest Euclidean norm across all the solutions on the line ℓ .

Hint: Pythagoras theorem may be helpful here.

Solution.

- (i) Let $\theta_t = (\theta_{t,1}, \theta_{t,2})$ refer to the entries of θ_t . Then, from the previous subpart we have that $\theta_{t,1} = 2\theta_{t,2}$ for all t . Plugging this into the equation for our solution set ℓ , we find that $\hat{\theta}_{\text{GD}} = (\frac{4}{5}, \frac{2}{5})$.
- (ii) Any $\hat{\theta} \in \ell$ is expressed as

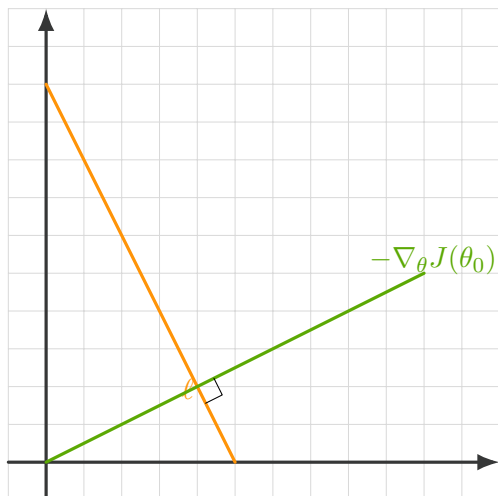
$$\begin{bmatrix} \hat{\theta}_1 \\ 2 - 2\hat{\theta}_1 \end{bmatrix}$$

Then we can write

$$\langle \hat{\theta} - \hat{\theta}_{\text{GD}}, \hat{\theta}_{\text{GD}} \rangle = \begin{bmatrix} \hat{\theta}_1 - \frac{4}{5} \\ 2 - 2\hat{\theta}_1 - \frac{2}{5} \end{bmatrix}^\top \begin{bmatrix} 4/5 \\ 2/5 \end{bmatrix} = 0$$

This concludes that $\hat{\theta} - \hat{\theta}_{\text{GD}} \perp \hat{\theta}_{\text{GD}}$.

- (iii) From the figure below, we see that $\|\hat{\theta}\|$ is the hypotenuse of a right triangle, whereas $\|\hat{\theta}_{\text{GD}}\|$ is the base of the triangle. Hence, by Pythagoras theorem, $\|\hat{\theta}\|$ must be larger.



Problem 3. (Gradient Descent for Linear Regression: Generalize from Simple Example.) Now, we generalize the argument from the simple example to general underdetermined systems. Assume full row-rank.

- (i) Show that for $\theta_0 = 0$, the iterates are always in the span by the rows of X . (**Hint:** you can use induction to show this).
- (ii) Show that $\hat{\theta}_{\text{GD}} = \hat{\theta}_{\text{UD}}$

This result is surprising and nontrivial. It states that amongst infinitely many solutions that attain the optimal objective value, gradient descent chooses the one with the minimum norm, which is a nice desirable property. This phenomenon, known as *implicit regularization*, helps explain the success of using gradient-based methods to train overparameterized models, like deep neural networks; this has been a very active area of research over the last 5 years!

Solution.

- (i) We prove the statement using induction. Write the update as

$$\theta_{t+1} = (I - \eta X^\top X)\theta_t + \eta X^\top y.$$

First is the base case $t = 0$. It follows from the assumption that $\theta_0 = 0$. If $\theta_t \in \mathcal{R}(X^\top)$, then $X^\top X\theta_t \in \mathcal{R}(X^\top)$ and so $(I - \eta X^\top X)\theta_t \in \mathcal{R}(X^\top)$; also $X^\top y \in \mathcal{R}(X^\top)$. By induction from $\theta_0 = 0 \in \mathcal{R}(X^\top)$,

$$\boxed{\theta_t \in \mathcal{R}(X^\top) \text{ for all } t \geq 0.}$$

- (ii) From above, we infer that there must exist some v such that $\hat{\theta}_{\text{GD}} = X^\top v$. Further from the definition of convergence, it must be true that

$$\nabla_\theta J(\hat{\theta}_{\text{GD}}) = 0 \implies X^\top X\hat{\theta}_{\text{GD}} - 2X^\top y = 0 \implies X^\top (X\hat{\theta}_{\text{GD}} - y) = 0 \implies X\hat{\theta}_{\text{GD}} - y = 0$$

The last implication follows since X is full row-rank. Then plugging $\hat{\theta}_{\text{GD}} = X^\top v$ into the above, we can solve for v to obtain $v = (XX^\top)^{-1}y$ and thus $\hat{\theta}_{\text{GD}} = Xv = \hat{\theta}_{\text{UD}}$.

Problem 4. (Step Size Choice in Gradient Descent.)

Consider minimizing a smooth quadratic function

$$f(\theta) = \frac{1}{2} \|X\theta - y\|_2^2$$

where $X \in \mathbb{R}^{m \times n}$ and $y \in \mathbb{R}^m$. Gradient descent updates are given by

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t), \quad \text{with } \nabla f(\theta) = X^\top (X\theta - y).$$

- (a) **Write the iteration in matrix form.** Show that the error $e_t = \theta_t - \theta^*$, where θ^* is the least-squares solution, evolves as

$$e_{t+1} = (I - \eta X^\top X)e_t.$$

Solution.

The least-squares solution satisfies $X^\top X\theta^* = X^\top y$. Subtracting this from the update gives

$$e_{t+1} = \theta_{t+1} - \theta^* = (\theta_t - \theta^*) - \eta X^\top X(\theta_t - \theta^*) = (I - \eta X^\top X)e_t.$$

- (b) **Condition for convergence.** Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $X^\top X$. Show that gradient descent converges for any initialization if and only if

$$|1 - \eta\lambda_i| < 1 \quad \text{for all } i.$$

Hence derive the allowable range of step sizes η . Hint: Write $X^\top X = Q\Lambda Q^\top$ with eigenvalues $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then set $e_t = Qz_t$ and show that $z_{t+1} = (I - \eta\Lambda)z_t$.

Solution. Write $X^\top X = Q\Lambda Q^\top$ with eigenvalues $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then $e_t = Qz_t$ evolves as

$$z_{t+1} = (I - \eta\Lambda)z_t, \quad \text{i.e.} \quad z_{t+1,i} = (1 - \eta\lambda_i)z_{t,i}.$$

Convergence requires $|1 - \eta\lambda_i| < 1$ for all i , i.e.

$$-1 < 1 - \eta\lambda_i < 1 \implies 0 < \eta < \frac{2}{\lambda_{\max}}.$$

- (c) **Optimal constant step size.** Within the convergent range, find the value of η that minimizes the worst-case contraction factor

$$\rho(\eta) = \max_i |1 - \eta\lambda_i|.$$

Show that the optimal choice satisfies

$$\eta^* = \frac{2}{\lambda_{\max} + \lambda_{\min}}$$

and compute the corresponding convergence rate ρ^* .

Solution.

The spectral radius is $\rho(\eta) = \max_i |1 - \eta\lambda_i|$. Since the function $|1 - \eta\lambda|$ is linear on $\lambda_{\min}$ and $\lambda_{\max}$, the optimal η equalizes the magnitudes at the two endpoints:

$$1 - \eta\lambda_{\min} = -(1 - \eta\lambda_{\max}) \quad \Rightarrow \quad \eta^* = \frac{2}{\lambda_{\max} + \lambda_{\min}}.$$

The resulting rate is

$$\rho^* = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\kappa - 1}{\kappa + 1},$$

where $\kappa = \lambda_{\max}/\lambda_{\min}$ is the condition number.

- (d) **Discussion.** Interpret intuitively what happens when 1. η is too small, and 2. η is too large. Relate your observations to the shape (conditioning) of the quadratic bowl $f(\theta)$.

Solution.

If η is *too small*, progress per step is slow — the algorithm takes many small steps down the bowl. On the other hand, if η is *too large*, updates overshoot and oscillate; once $\eta > 2/\lambda_{\max}$, the iterates diverge. Poor conditioning ($\kappa \gg 1$) makes it hard to pick a single η that works well for all curvature directions.